Customer Support Ticket Categorization and Prioritization Using Natural Language Processing

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Abstract

Efficient and effective customer service relies heavily on the systematic management of customer support tickets. This paper presents an Natural Language Processing (NLP) - driven system designed to automate the categorization and prioritization of customer support tickets, significantly enhancing service efficiency. The system analyzes ticket content using machine learning models and deep learning models for both categorization and prioritization tasks. Categorization is achieved using Topic Modeling (NMF) to identify department-specific categories, while ticket priority levels are determined by extracting urgency and impact keywords. By leveraging these techniques, the system streamlines ticket handling, reduces manual intervention, and optimizes resource allocation. Experimental results demonstrate high accuracy, scalability, and improved operational efficiency, ultimately enhancing customer satisfaction.

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